

A cubic-box counterexample to Conjecture 425

Local AI-assisted solution draft
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Source and interpretation. Conjecture 00000000425 in The Last Math Competition, at commit `efab34b80a63963991d6c7ed625442a89a328a44`, asserts that the coefficient sequence in the diagonal specialization of the boxed plane partition generating function is always log-concave. We use the standard volume generating polynomial, with the diagonal specialization meaning equal box side lengths. The counterexample is the $2 \times 2 \times 2$ box.

Definitions. A plane partition in this box is an array $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with entries in $\{0, 1, 2\}$, weakly decreasing along both rows and columns: $a \geq b$, $a \geq c$, $b \geq d$, and $c \geq d$. Its volume is $a + b + c + d$. Let c_k count such arrays of volume k . Log-concavity requires $c_k^2 \geq c_{k-1}c_{k+1}$ at every interior index.

Counterexample and proof. There is exactly one array of volume zero, namely the zero array, and exactly one of volume one, with a single 1 in the top-left entry. The three arrays of volume two are

$$\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}.$$

These are exhaustive: either one entry equals 2, which by monotonicity must be the top-left entry, or two entries equal 1, giving exactly the two displayed order ideals. Consequently

$$c_0 = 1, \quad c_1 = 1, \quad c_2 = 3, \quad c_1^2 = 1 < 3 = c_0 c_2.$$

Thus the asserted universal log-concavity is false. It is unnecessary to consider the additional equality characterization after this first clause has failed.

Independent polynomial check. The complete polynomial, obtained both by direct enumeration and by the MacMahon product, is

$$\prod_{i,j,k=1}^2 \frac{1 - q^{i+j+k-1}}{1 - q^{i+j+k-2}} = 1 + q + 3q^2 + 3q^3 + 4q^4 + 3q^5 + 3q^6 + q^7 + q^8.$$

The accompanying Python verifier independently computes the product as an exact integer formal power series and enumerates the arrays.

Formal verification and scope. The self-contained Lean 4.19.0 file `Main.lean` defines all four entries as `Fin 3`, proves the 81-array enumeration is complete and has no duplicates, proves the filtered membership predicate is exactly the plane-partition condition, and computes every coefficient. The theorem `Conjecture425.conjecture425_counterexample` proves the negation of log-concavity for this genuine diagonal box. All calculations use kernel reduction; there are no proof placeholders, custom axioms, or native decision procedures. The formal proof follows the direct counting definition; the MacMahon identity is an independent check, not an assumed Lean premise.

Alternative reading of “coefficientwise.” If the intended claim instead compares consecutive polynomials $F_n(q)$ for $n \times n \times n$ boxes, it is false as well. The completely filled box is the unique term of maximum volume, so F_n is monic of degree n^3 . At $n = 2$, the polynomial $F_2(q)^2 - F_1(q)F_3(q)$ has coefficient -1 at q^{28} , because the two product degrees are 16 and 28. This additional argument covers that reading in ordinary mathematics; the Lean artifact formalizes the scalar coefficient-sequence reading above.